

A Hybrid Ensemble Framework for Intrusion Detection and Attack Classification Using Random Forest, Kernel PCA–K-Means, and XGBoost

Group 2 - Kalyan Brata Majumder, Karthik Mekala and Kunal Gurtoo

Supervisor - Professor Anamika Singh

Department of Applied Artificial Intelligence, University of San Diego

Github Link-

<https://github.com/networkenthusiast/Hybrid-Ensemble-Intrusion-Detection-and-Attack-Classification.git>

Acknowledgement

We would like to express our sincere gratitude to our project guide, **Prof. Anamika Singh**, for her invaluable guidance, constructive feedback, and continuous support throughout the course of this project.

We would also like to acknowledge the collaborative efforts of our team members.

Kalyan Brata Majumder extensively contributed to this project in designing and developing the hybrid intrusion detection framework. He implemented the binary Random Forest classifier for attack detection, performed feature selection using feature importance analysis, and developed the Kernel PCA + K-Means clustering model to improve unsupervised traffic classification. He integrated supervised and unsupervised outputs into a stacked ensemble architecture and designed the final XGBoost-based hybrid model. He conducted model evaluation, comparative analysis, feature engineering, hyperparameter tuning, and performance optimization to improve intrusion detection accuracy and robustness. Additionally, he created the technical report for this project.

Karthik Mekala significantly contributed in identifying and grouping attack categories in a manner that reduced class imbalance and improved the effectiveness of multiclass classification. He also developed the attack category encoding framework, enabling downstream machine learning models to train and predict effectively. Additionally, he created the baseline **PCA + K-Means** model, which served as an important benchmark for evaluating the proposed **Kernel PCA + K-Means** approach.

Kunal Gurtoo contributed to the attack-type classification component of this project. He played a key role in designing the multiclass prediction framework and incorporating attack-category probability outputs into the hybrid ensemble architecture using Softmax. His work enabled the model to leverage attack-specific probability distributions as additional intelligence for the final XGBoost classifier, improving both predictive capability and attack attribution within the intrusion detection system. Additionally, he created the presentation for this project.

Finally, we would like to acknowledge the availability of the CICIDS2017 dataset and related resources, which provided a strong foundation for conducting this research. The collective effort, collaboration, and knowledge sharing among team members have been essential in successfully completing this project.

Contents

1. Introduction	6
2. Literature Review	6
3. Data Summary	7
4. Methodology	12
4.1 Model 1: Supervised Learning Approach (Random Forest)	13
4.2 Model 1A : Multiclass Attack Type Classification Using Random Forest	17
4.3 Unsupervised Feature Engineering Using K-Means Clustering	22
4.4 Model 2 : Unsupervised Learning Model: Kernel PCA+K-Means Clustering	26
4.5 Model 3 : Final Hybrid Ensemble Model	33
5. Overall Assessment and Conclusion	38
6. Deployment Strategy	38
7. Limitations and Future Scope	39
8. References	39

List Of Figures

S.No.	Figure No.	Figure Caption	Page No.
1.	3.3	<i>Distribution of network table labels</i>	8
2.	3.4	<i>Correlation Features</i>	9
3.	3.6.3	<i>Attack Category Distribution</i>	11
4.	4	<i>System Architecture of the Proposed Model</i>	12

5.	4.1.2.1	<i>Classification Report (Supervised Model-Random Forest)</i>	14
6.	4.1.2.2	<i>Confusion Matrix (Supervised Model-Random Forest)</i>	14
7.	4.1.2.3	<i>ROC Curve (Supervised Model-Random Forest)</i>	15
8.	4.1.2.4	<i>Top 15 Feature Importance (Supervised Model-Random Forest)</i>	15
9.	4.1.2.5	<i>Learning Curve (Supervised Model-Random Forest)</i>	16
10.	4.1.2.6	<i>Meta Features (Supervised Model-Random Forest)</i>	16
11.	4.2.2.5	<i>Softmax-style Probability Generation (Supervised Model- Softmax)</i>	18
12.	4.2.2.6	<i>Meta Feature Generation (Supervised Model- Softmax)</i>	18
13.	4.2.2.7	<i>Classification Report (Supervised Model- Softmax)</i>	19
14.	4.2.2.8	<i>Confusion Matrix Analysis (Supervised Model- Softmax)</i>	20
15.	4.2.2.9	<i>ROC-AUC Curve Analysis (Supervised Model- Softmax)</i>	20
16.	4.2.2.10	<i>Learning Curve Analysis (Supervised Model- Softmax)</i>	21
17.	4.3.2	<i>K- Means Clustering in PCA Space</i>	23
18.	4.4.3	<i>Elbow Method and Silhouette Analysis (Unsupervised Model Kernel PCA+K-Means)</i>	27
19.	4.4.4	<i>Cluster Interpretation (Unsupervised Model Kernel PCA+K-Means)</i>	28
20.	4.4.6	<i>Classification Performance (Unsupervised Model Kernel PCA+K-Means)</i>	28
21.	4.4.7	<i>Confusion Matrix Analysis (Unsupervised Model Kernel PCA+K-Means)</i>	28
22.	4.4.8	<i>ROC-AUC Analysis (Unsupervised Model Kernel PCA+K-Means)</i>	29

23	4.4.13	<i>Classification Performance (Unsupervised Model Kernel PCA+K-Means)</i>	30
24	4.4.14	<i>Confusion Matrix Comparison(Unsupervised Model Kernel PCA+K-Means)</i>	31
25	4.4.15	<i>Visual Cluster Separation (Unsupervised Model Kernel PCA+K-Means)</i>	31
26	4.4.16	<i>Meta Feature Separation (Unsupervised Model Kernel PCA+K-Means)</i>	32
27			33
28	4.5.2	<i>Hybrid Architecture (Final Ensemble Model)</i>	35
29	4.5.7.1	<i>Classification Report (Final Ensemble Model)</i>	36
30	4.5.7.2	<i>Confusion Matrix Analysis (Final Ensemble Model)</i>	36
31	4.5.7.3	<i>ROC-AUC Ensemble Model (Final Ensemble Model)</i>	37
32	4.5.7.4	<i>Learning Curve Analysis (Final Ensemble Model)</i>	38
	4.5.7.5	<i>Feature Importance Analysis (Final Ensemble Model)</i>	

1. INTRODUCTION

Modern network environments generate massive volumes of traffic, making the identification of malicious activities increasingly challenging. While detecting whether traffic is benign or malicious remains a fundamental requirement of Intrusion Detection Systems (IDS), security analysts also require information about the nature of an attack to support effective incident response and threat mitigation.

This project proposes a **Hybrid Ensemble Intrusion Detection and Attack Classification Framework** that combines supervised, unsupervised, and ensemble learning techniques to improve network security. The framework integrates **Random Forest-based intrusion detection**, **Kernel PCA + K-Means clustering**, and **multiclass attack categorization** to identify both malicious traffic and specific attack families. The outputs of these models are combined within an **XGBoost ensemble classifier**, enabling the system to leverage traffic behavior, attack-type intelligence, and structural network patterns. Using the **CICIDS2017 dataset**, the proposed approach provides accurate intrusion detection along with attack attribution capabilities for enhanced cybersecurity operations.

2. LITERATURE REVIEW

The CICIDS2017 dataset has emerged as a prominent benchmark for evaluating machine learning-based intrusion detection systems due to its realistic traffic profiles and diverse attack scenarios [1]. Supervised algorithms, particularly Random Forest and XGBoost, have demonstrated strong classification performance, while feature selection and dimensionality reduction techniques have been employed to enhance efficiency and mitigate redundancy [2], [4], [6], [7].

Recent research has increasingly focused on hybrid architectures that integrate clustering and classification to capture both explicit and latent traffic characteristics [3]. Nonlinear transformation methods such as Kernel PCA have further improved data representation by preserving complex network behaviors that may not be captured through conventional linear techniques [8]. In parallel, ensemble learning strategies have shown effectiveness in improving predictive robustness through the integration of multiple learning paradigms [5], [10].

Despite these advances, existing studies predominantly emphasize binary intrusion detection. This work extends current research by combining attack categorization, nonlinear clustering, and ensemble learning within a unified framework, thereby enhancing both detection capability and attack attribution [9].

3. DATA SUMMARY

3.1 Dataset Description

- [1], The study uses the **CICIDS2017 dataset** from Kaggle, containing labelled network traffic with multiple attack types and 78 flow-based features suitable for machine learning.
- The dataset was generated in a realistic testbed environment at the Canadian Institute for Cybersecurity, capturing both benign and malicious traffic.
- Data collection spans five days, including one fully benign day and multiple days with attacks such as DoS, DDoS, brute force, web attacks, botnet, and infiltration.
- The dataset reflects modern network behaviour and is widely used for intrusion detection research as a replacement for older datasets.
- Some limitations include class imbalance, label noise, and redundant or irrelevant flows, which require preprocessing.

3.2 Data Pre-Processing

- Cleaning negative values – Negative or invalid values were handled by removing a few records and replacing others with zero to avoid excessive data loss.
- Column consistency check – Ensured all CSV files had identical column structure before merging to prevent data misalignment.
- Removal of infinite values – Infinite values arising from division-based features were identified and removed to ensure stable model training.
- Duplicate column removal – Redundant or duplicate columns were eliminated to improve data quality and reduce dimensionality.

3.3 Major challenges in Network Distribution Labels

The figure shows a highly imbalanced dataset, where BENIGN traffic overwhelmingly dominates. Among attacks, DoS Hulk, Port Scan, and DDoS appear more frequently, while others like Heartbleed, SQL Injection, Infiltration, and Web Attacks have very few samples.

This imbalance can bias the model toward predicting the BENIGN class. Rare attack types may not be effectively learned by the model.

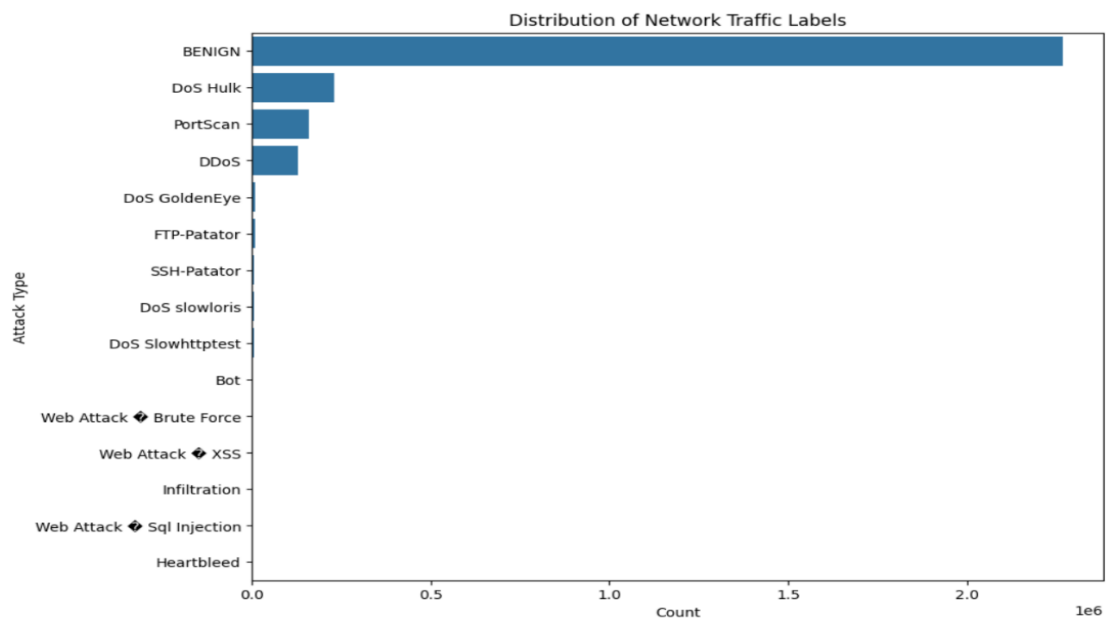


Fig 3.3: Distribution of network table labels

3.4 Correlation Analysis of Numerical Features

The analysis identified several highly correlated and redundant features. Removing them reduced dimensionality, improved training efficiency, minimized overfitting, and enhanced interpretability.

Overall, correlation-based feature selection which we will see in the following section, streamlined the dataset while retaining important information, resulting in a more efficient and robust feature set for model training.

A Hybrid Ensemble Framework for Intrusion Detection and Attack Classification Using Random Forest, Kernel PCA- K-Means, and XGBoost

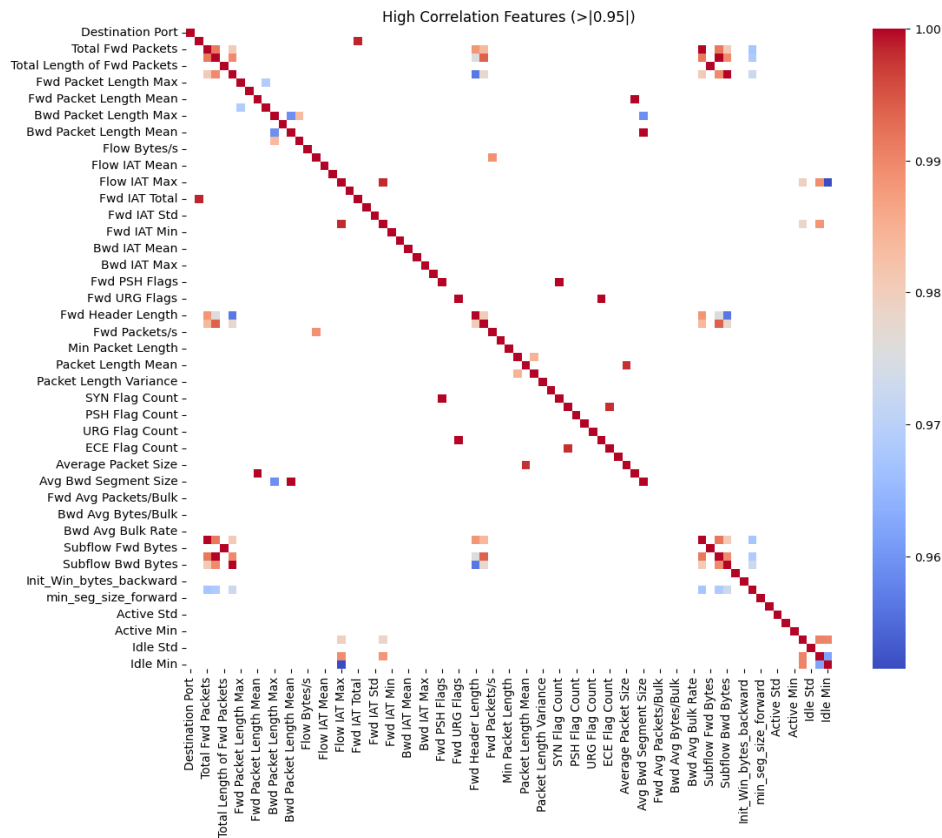


Fig 3.4: Correlation Features

3.5 Binary Label Encoding

The dataset is converted into a binary classification problem by encoding BENIGN traffic as 0 and all attack types as 1. Labels are first standardized (trimmed and uppercased) for consistency. This simplifies the task and aligns with practical intrusion detection systems.

3.6 Attack Category Consolidation and Encoding

3.6.1 Objective

To reduce class fragmentation and create a more meaningful multiclass classification problem, the original CICIDS2017 attack labels were consolidated into six high-level attack categories and encoded into numerical values using LabelEncoder.

3.6.2 Attack Category Encoding

Original attack labels were mapped to their corresponding attack categories and subsequently converted into numerical values using Scikit-learn's LabelEncoder. The encoded labels were used as target variables for multiclass attack classification.

Attack Category	Encoded Class
Benign	0
Credential_Attack	1
DoS_DDoS	2
Malware_Compromise	3
Reconnaissance	4
Web_Attack	5

Table 3.6.2 – Attack Category Encoding

3.6.3 Category Distribution

Following consolidation, the dataset distribution was:

Attack Category	Count	Percentage
Benign	2,268,589	80.31%
DoS_DDoS	379,554	13.44%
Reconnaissance	158,804	5.62%
Credential_Attack	15,333	0.54%
Malware_Compromise	1,998	0.07%
Web_Attack	673	0.02%

Table 3.6.3 – Attack Category Distribution

Total records: 2,825,951

3.6.3 Distribution Analysis

Figure illustrates the distribution of attack categories after excluding benign traffic. The DoS_DDoS category dominates the attack dataset, followed by Reconnaissance traffic. In contrast, Credential_Attack, Malware_Compromise, and Web_Attack categories contain significantly fewer samples, indicating substantial class imbalance. Despite this imbalance, the

consolidated taxonomy provides a more interpretable attack representation while reducing the complexity of multiclass intrusion detection.

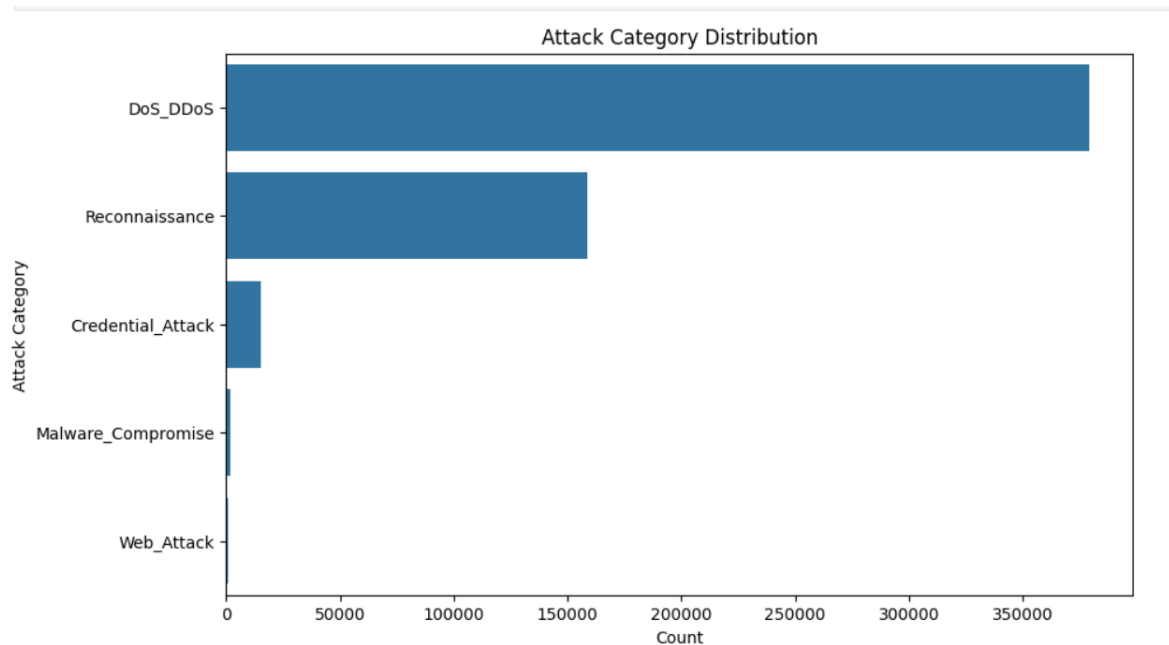


Fig 3.6.3 – Attack Category Distribution

3.6.4 Key Outcome

The encoded attack categories were used to train a multiclass attack classification model. The predicted attack classes and their corresponding probability scores were subsequently incorporated as meta-features in the final hybrid intrusion detection framework.

3.7 Train–Test Split and Class Imbalance Handling

The dataset is imbalanced, with benign traffic dominating attack instances. Stratified sampling is used to split the data while preserving class distribution.

Under sampling is applied only to the training set to balance classes and avoid data leakage. This ensures improved detection of attack traffic while maintaining realistic evaluation conditions.

4. METHODOLOGY

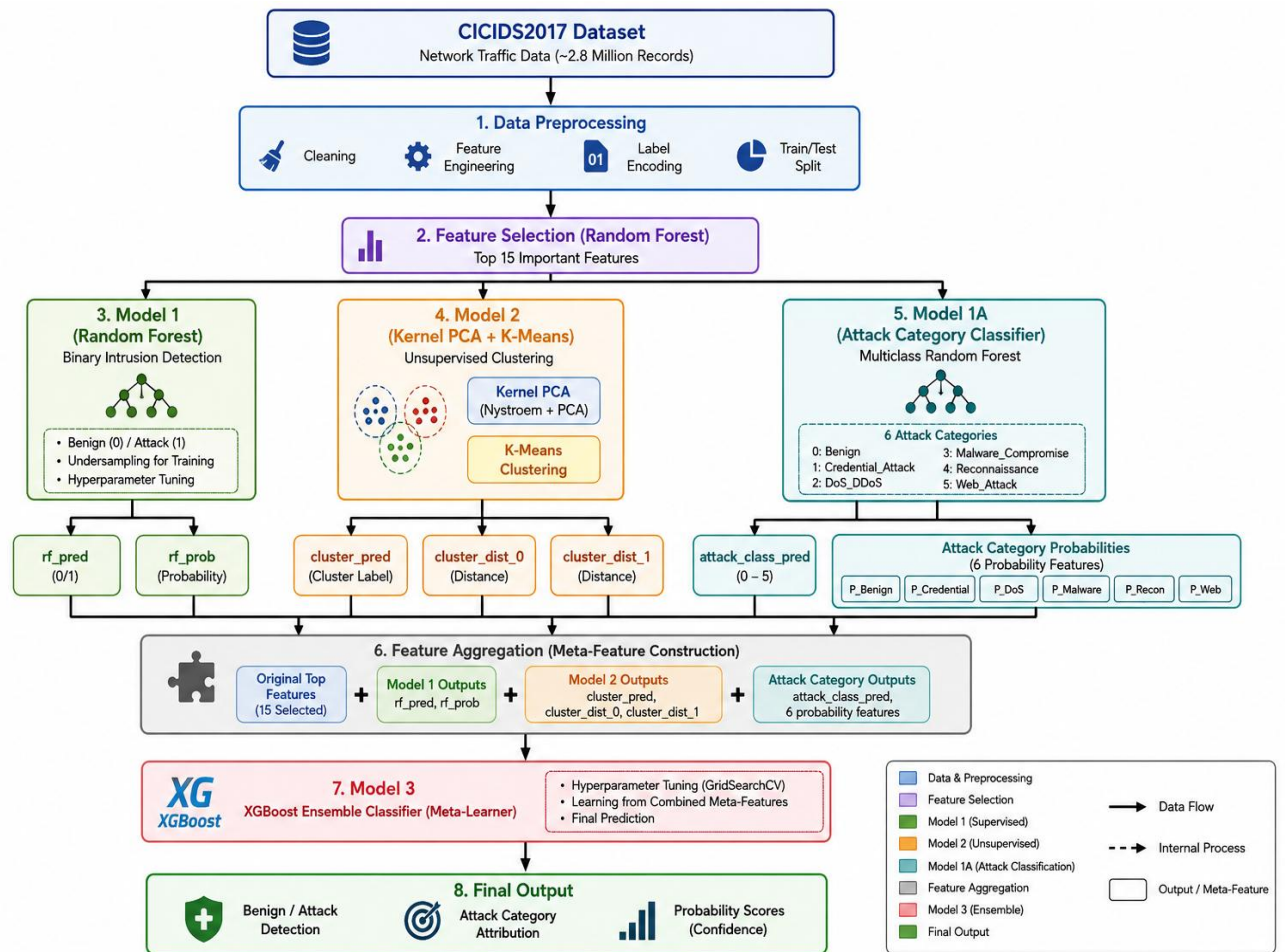


Fig 4: System Architecture of the Proposed Model

The proposed framework combines supervised, unsupervised, and ensemble learning techniques for network intrusion detection. Random Forest, Kernel PCA + K-Means, and a multiclass attack-category classifier generate complementary prediction and probability features, which are aggregated and provided to an XGBoost meta-learner. This hybrid architecture enables accurate attack detection, attack-family identification, and improved decision confidence.

4.1 MODEL 1: SUPERVISED LEARNING APPROACH(RANDOM FOREST)

4.1.1 SYSTEM OVERVIEW

Model 1 uses a supervised learning approach to classify network traffic as normal or malicious using labelled data. It follows a pipeline of preprocessing, feature engineering, training, and evaluation. A Random Forest classifier is applied on the CICIDS2017 dataset, with performance improved through feature selection, hyperparameter tuning, and evaluation.

4.1.1.1 Data Preparation

The dataset was split into features (X) and target (y), where *Traffic_Label* represents the binary class. A balanced training dataset was used to handle class imbalance and ensure fair learning across both classes.

4.1.1.2 Model Selection

A Random Forest classifier was chosen for its ability to handle high-dimensional data, resist overfitting, and provide feature importance. It was configured with `random_state=42` for reproducibility and `n_jobs=-1` for parallel processing.

4.1.1.3 Hyperparameter Tuning

Grid Search with 5-fold Stratified Cross-Validation identified optimal parameters: `max_depth=20`, `min_samples_leaf=2`, `min_samples_split=5`, and `n_estimators=100`. The top 15 features were selected based on random forest feature importance to improve efficiency and interpretability.

4.1.1.4 Model Retraining

The model was retrained using the selected top features, with `X_train_top` and `X_test_top` as the training and testing sets. This enables the model to focus on the most relevant information.

4.1.2 MODEL EVALUATION AND RESULTS

4.1.2.1 Classification Report

The model was evaluated using standard classification metrics, including accuracy, precision, recall, and F1-score. The results indicate strong overall performance, with an accuracy of approximately 99% and high precision and recall across both classes.

Classification Report:					
	precision	recall	f1-score	support	
0	1.00	0.99	1.00	453718	
1	0.98	1.00	0.99	111273	
accuracy			0.99	564991	
macro avg	0.99	1.00	0.99	564991	
weighted avg	0.99	0.99	0.99	564991	

Fig 4.1.2.1: Classification Report (Supervised Model-Random Forest)

The model achieves strong and balanced performance with F1-scores of 0.99.

- Benign (Class 0): High precision (1.00) and recall (0.99) indicate accurate classification with minimal false positives. Attack (Class 1): Perfect recall (1.00) ensures almost all attacks are detected, with high precision (0.98).

Interpretation:

The model effectively distinguishes between normal and malicious traffic, maintaining a strong precision–recall balance, with high attack detection and improved efficiency through feature selection.

4.1.2.2 CONFUSION MATRIX ANALYSIS

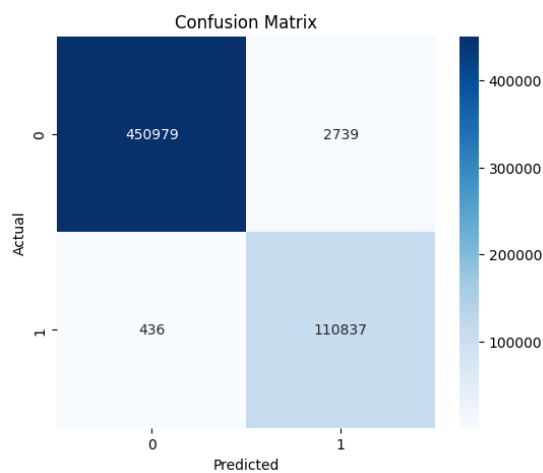


Fig 4.1.2.2: Confusion Matrix (Supervised Model-Random Forest)

The Random Forest model shows strong performance with high true negatives (450,979) and true positives (110,837), indicating effective detection of both benign and malicious traffic. False positives (2,739) are low, while false negatives (436) are minimal, ensuring most attacks are correctly identified.

Overall, the model demonstrates reliable intrusion detection performance, with low misclassification rates and strong emphasis on detecting malicious activity.

4.1.2.3 ROC CURVE ANALYSIS

The ROC curve shows excellent model performance, with an AUC of 0.9996, indicating near-perfect class separation. The curve’s proximity to the top-left corner reflects a high true positive rate and low false positive rate. This confirms the model’s strong ability to distinguish between benign and malicious traffic, making it highly effective for intrusion detection, with further validation recommended for generalization.

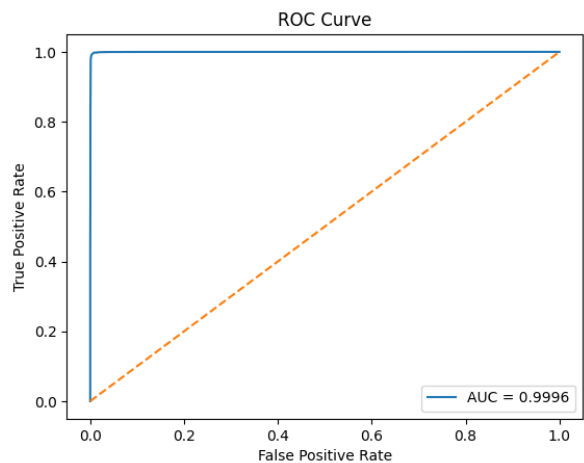


Fig 4.1.2.3: ROC Curve (Supervised Model--Random Forest)

4.1.2.4 FEATURE IMPORTANCE ANALYSIS

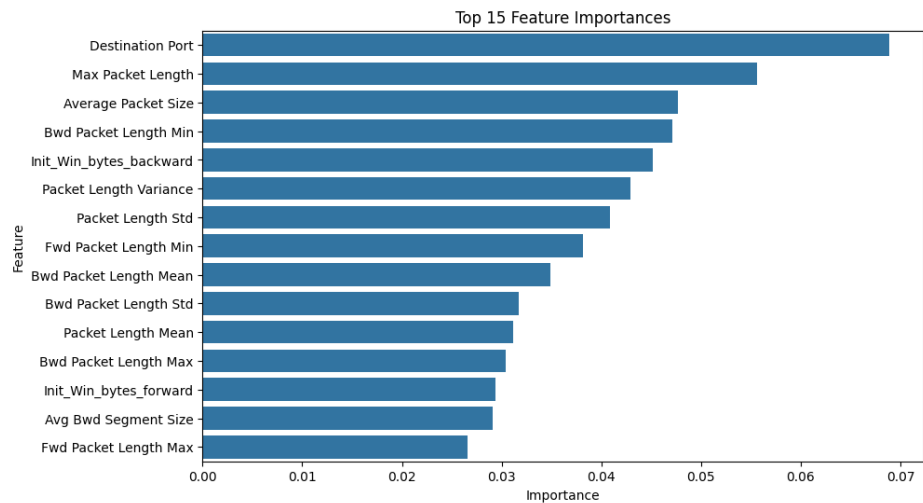


Fig 4.1.2.4: Top 15 Feature Importance (Supervised Model--Random Forest)

Random Forest feature importance identifies key features such as Destination Port, packet length metrics, and forward/backward traffic characteristics, along with TCP window size features for connection-level insights. Overall, traffic patterns are crucial for attack detection. Selecting top features improves efficiency, reduces dimensionality, and enhances interpretability while maintaining performance.

4.1.2.5 LEARNING CURVE ANALYSIS

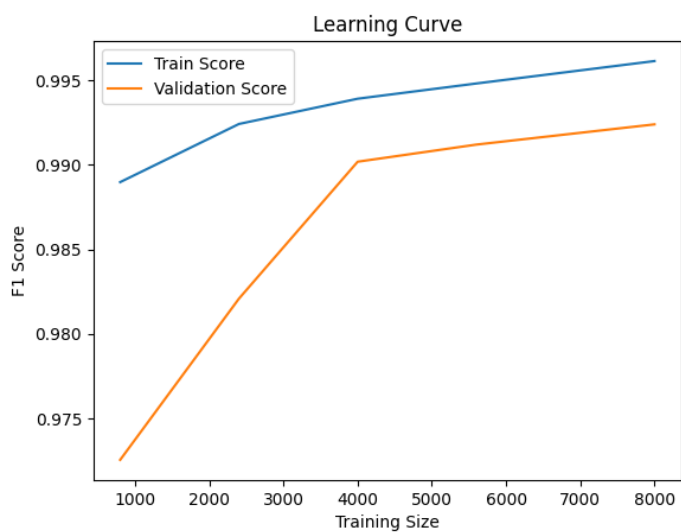


Fig 4.1.2.5: Learning Curve (Supervised Model--Random Forest)

The model shows strong performance, with training F1-score increasing from **0.989 to 0.996** and validation F1-score from **0.972 to 0.992** as data size grows. The small gap (~0.003–0.004) indicates minimal overfitting and good generalization, showing a stable and well-balanced model.

4.1.2.6 Meta Feature

	Destination Port	Flow Duration	Total Fwd Packets	Total Backward Packets	Total Length of Fwd Packets	Total Length of Bwd Packets	Fwd Packet Length Max	Fwd Packet Length Min	Fwd Packet Length Mean	Fwd Packet Length Std	...	Active Mean	Active Std	Active Max	Active Min	Idle Mean	Idle Std	Idle Max	Idle Min	rf_pred	rf_prob
2731287	53	161	2	2	62	94	31	31	31.00000	0.000000	...	0.0	0.0	0	0	0.0	0.0	0	0	0	0.000000
2803437	443	3	2	0	12	0	6	6	6.00000	0.000000	...	0.0	0.0	0	0	0.0	0.0	0	0	0	0.042055
1920623	80	5259727	4	4	437	839	437	0	109.25000	218.500000	...	0.0	0.0	0	0	0.0	0.0	0	0	0	0.000000
644010	443	3	2	0	12	0	6	6	6.00000	0.000000	...	0.0	0.0	0	0	0.0	0.0	0	0	0	0.042055
1818676	443	191849	32	49	871	64661	389	0	27.21875	83.719476	...	0.0	0.0	0	0	0.0	0.0	0	0	0	0.000000

Fig 4.1.2.6: Meta Feature (Supervised Model--Random Forest)

The Binary Random Forest model generates **rf_pred** (predicted class label) and **rf_prob** (attack probability), providing supervised attack intelligence that is incorporated into the final XGBoost ensemble model.

4.2 MODEL 1A : MULTICLASS ATTACK TYPE CLASSIFICATION USING RANDOM FOREST

4.2.1 SYSTEM OVERVIEW:

After developing a binary intrusion detection model to distinguish between benign and malicious traffic, a second supervised learning model was developed to identify the specific category of attack associated with each malicious network flow. The objective of this model is to classify network traffic into one of six attack categories. In addition to attack-type prediction, the class probabilities generated by the model are used as meta-features for the final ensemble model (Model 3).

4.2.2 METHODOLOGY

4.2.2.1 Training Dataset

To ensure consistency with the binary Random Forest model, the same balanced training dataset was reused. The attack category labels corresponding to the selected training records were extracted. This guarantees that both binary and multiclass models are trained on identical network flows.

4.2.2.2 Feature Selection

The same top 15 features identified through Random Forest feature importance analysis were reused.

4.2.2.3 Random Forest Model

A multiclass Random Forest classifier was trained using **200 trees, maximum depth of 20, minimum split size of 5, and minimum leaf size of 2** (random_state=42). The model was designed to classify network traffic into six attack categories: **Benign, Credential_Attack, DoS_DDoS, Malware_Compromise, Reconnaissance, and Web_Attack**.

4.2.2.4 Attack Type Prediction

The trained model was evaluated on the unseen test dataset and generated an attack category prediction for every network flow. The predicted attack category (attack_class_pred) was stored as a meta-feature within the Model 3 dataset. It provides attack-family information in addition to binary attack detection. In addition to that, it enables the final ensemble model to understand the nature of the detected attack and improves contextual awareness during final classification.

4.2.2.5 Softmax-Style Probability Generation

Although Random Forest does not use a Softmax output layer internally, the `predict_proba()` function generates class-membership probabilities that behave similarly to Softmax probabilities. The model generated a probability matrix of size **564,991 × 6**, representing the likelihood of each network flow belonging to one of the six attack categories. For example, a prediction of **[0.98, 0.00, 0.01, 0.00, 0.01, 0.00]** indicates a **98% probability of Benign traffic**, with small probabilities assigned to **DoS_DDoS** and **Reconnaissance** categories.

Benign	Credential	DoS_DDoS	Malware	Recon	Web
0.98	0.00	0.01	0.00	0.01	0.00

Fig 4.2.2.5: Softmax-Style Probability Generation (Supervised Model—Softmax)

4.2.2.6 Meta-Feature Generation

The probability associated with each attack category was stored as a separate feature.

Generated Features- `attack_prob_Benign`, `attack_prob_Credential_Attack`, `attack_prob_DoS_DDoS`, `attack_prob_Malware_Compromise`, `attack_prob_Reconnaissance` and `attack_prob_Web_Attack`. These probability features provide richer information than a single predicted class and allow the final XGBoost model to utilize attack-category confidence scores during decision making.

rf_pred	rf_prob	attack_class_pred	attack_prob_Benign	attack_prob_Credential_Attack	attack_prob_DoS_DDoS	attack_prob_Malware_Compromise	attack_prob_Reconnaissance	attack_prob_Web_Attack
0	0.000000	0	1.000000	0.0	0.000000	0.0	0.000000	0.0
0	0.042055	0	0.987650	0.0	0.012329	0.0	0.000021	0.0
0	0.000000	0	1.000000	0.0	0.000000	0.0	0.000000	0.0
0	0.042055	0	0.987650	0.0	0.012329	0.0	0.000021	0.0
0	0.000000	0	0.993333	0.0	0.006667	0.0	0.000000	0.0

Fig 4.2.2.6: Meta-Feature Generation (Supervised Model—Softmax)

4.2.2.7 Classification Performance

Classification Report

Attack Class	Precision	Recall	F1 Score
Benign	1.00	1.00	1.00
Credential_Attack	0.97	0.90	0.93
DoS_DDoS	0.99	1.00	0.99
Malware_Compromise	0.59	0.74	0.66
Reconnaissance	0.99	0.99	0.99
Web_Attack	0.00	0.00	0.00

Fig 4.2.2.7: Classification Report (Supervised Model—Softmax)

Overall Accuracy ~100% Weighted and F1 Score: ~1.00

4.2.2.8 Interpretation of Results

Attack Category Performance Analysis

<u>Attack Category</u>	<u>Precision</u>	<u>Recall</u>	<u>F1 Score</u>	<u>Interpretation</u>
Benign	1.00	1.00	1.00	The model perfectly distinguished benign traffic from attack traffic.
DoS_DDoS	0.99	1.00	0.99	Denial-of-Service attacks exhibited strong and distinctive traffic patterns, resulting in near-perfect classification.
Reconnaissance	0.99	0.99	0.99	Reconnaissance activities such as port scanning were highly separable from other attack categories.
Credential_Attack	0.97	0.90	0.93	The majority of brute-force authentication attacks were correctly identified.
Malware_Compromise	0.59	0.74	0.66	Performance was moderate due to the limited number of available training samples.
Web_Attack	0.00	0.00	0.00	The model failed to learn a reliable decision boundary because this category represented less than 0.02% of the dataset.

Table 4.2.2.8: Attack Category Performance Analysis (Supervised Model—Softmax)

Overall Accuracy: ~100% Weighted F1 Score: ~1.00

4.2.2.8 Confusion Matrix Analysis

The confusion matrix demonstrates strong multiclass classification performance, with most **Benign**, **DoS_DDoS**, **Reconnaissance**, and **Credential_Attack** samples correctly identified. Misclassifications were mainly concentrated in the **Malware_Compromise** and **Web_Attack** categories due to their limited sample sizes, highlighting the impact of class imbalance on minority attack classes.

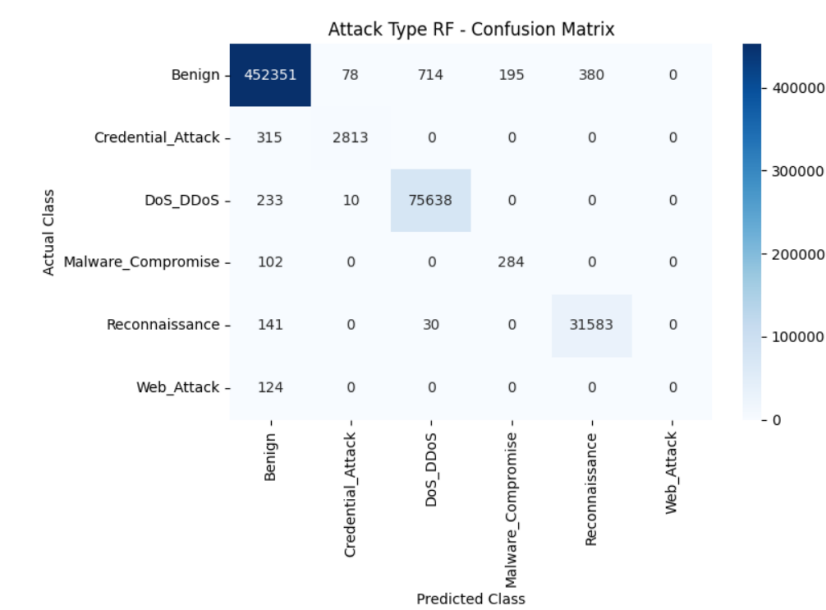


Fig 4.2.2.8 Confusion matrix Analysis (Supervised Model—Softmax)

4.2.2.9 ROC – AUC Curve Analysis

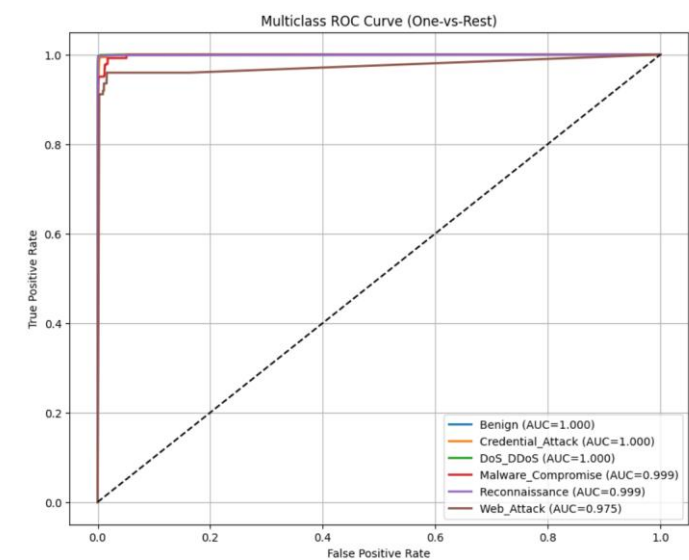


Fig 4.2.2.9: ROC – AUC Curve Analysis (Supervised Model—Softmax)

A One-vs-Rest (OvR) ROC analysis showed excellent performance across all attack categories, with **AUC values ranging from 0.975 to 1.000**. Even the minority **Web_Attack** class achieved an **AUC of 0.975**, indicating strong class separability and reliable probability estimation despite severe class imbalance.

4.2.2.10 Learning Curve Analysis

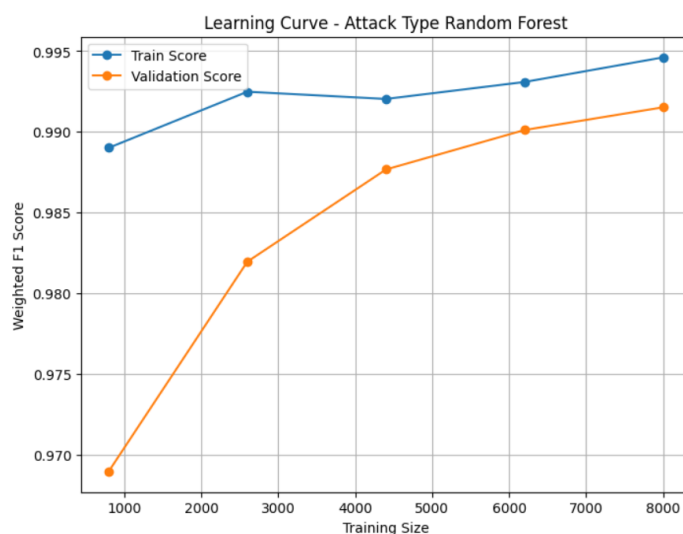


Fig 4.2.2.10: learning Curve Analysis (Supervised Model—Softmax)

The learning curve demonstrates strong generalization performance, with the training and validation F1 scores converging at approximately **0.995** and **0.992**, respectively. The close alignment between the two curves indicates minimal overfitting and low model variance. Furthermore, validation performance improves consistently as additional training samples are introduced, suggesting that the model effectively learns underlying attack patterns and maintains robust performance on unseen data.

4.2.3 Conclusion

The multiclass Random Forest model achieved excellent performance in identifying major attack categories, particularly **Benign**, **DoS_DDoS**, and **Reconnaissance** traffic, while demonstrating strong generalization and minimal overfitting. Although **Malware_Compromise** and **Web_Attack** remained challenging due to severe class imbalance, the model successfully generated attack-category predictions and probability scores. These outputs were integrated into Model 3 alongside binary Random Forest predictions and Kernel PCA + K-Means clustering features, providing valuable attack-

type intelligence that enhanced the final XGBoost ensemble model's detection accuracy and robustness.

4.3 UNSUPERVISED FEATURE ENGINEERING USING K-MEANS CLUSTERING

4.3.1 SYSTEM OVERVIEW

The goal of this step is to extract additional informative features from the dataset using an unsupervised learning approach. Specifically, we use K-Means clustering to identify inherent patterns in the data and convert them into new features that can improve downstream supervised models.

4.5.1.1 Balanced Sampling for Unsupervised Learning

In order to mitigate class imbalance, a balanced dataset was created by randomly selecting **50,000 benign** and **50,000 attack** samples. The combined dataset was then shuffled to ensure that K-Means learns representative patterns from both classes equally.

4.5.1.2 Data Preparation

The dataset is split into: Features (X_train_unsup) Labels (y_train_unsup). The same test set used in supervised learning is reused for consistency.

4.5.1.3 Feature Scaling

Standardization is applied using StandardScaler to normalize the feature space. This is important because: K-Means is distance-based . Features with larger scales would otherwise dominate clustering.

4.5.1.4 Dimensionality Reduction using PCA

PCA reduced the feature space from **77 to 20 features**, removing noise and redundancy while improving clustering efficiency and reducing computational complexity.

4.5.1.5 K-Means Clustering

MiniBatch K-Means was applied with **2 clusters**, a **batch size of 5,000**, and **10 initialization runs**. This approach was selected for its computational efficiency and ability to scale effectively to large network traffic datasets while maintaining clustering performance.

4.3.1.6 Cluster Interpretation

Since K-Means is an unsupervised algorithm, the generated cluster labels do not directly represent benign or attack traffic. To assign meaning to each cluster, the average attack rate within each cluster was calculated. Cluster 0 had an attack rate of 40.15% and was mapped to Benign, while Cluster 1 had an attack rate of 87.77% and was mapped to Attack. This mapping indicates that Cluster 1 predominantly contains malicious traffic, whereas Cluster 0 primarily represents benign network flows.

4.3.1.7 Feature Engineering from Clustering

Three clustering-based features were generated: `cluster_pred`, `cluster_dist_0`, and `cluster_dist_1`. These features capture cluster assignments and distances to cluster centroids, providing additional structural information that enhances the final ensemble model's classification capability.

4.3.1.8 Integration into Final Dataset

The clustering-derived features (`cluster_pred`, `cluster_dist_0`, and `cluster_dist_1`) were added to the final dataset alongside the original features and Random Forest outputs (`rf_pred` and `rf_prob`). This transforms unsupervised clustering information into useful meta-features for the hybrid ensemble model.

4.3.1.9 Conclusion

K-Means clustering successfully identifies underlying structure in the network traffic data. By converting cluster assignments and distances into features, we enhance the model's ability to distinguish between benign and malicious traffic. This step plays a crucial role in improving the performance of the final classification model.

4.3.2 Visualization of K-Means Clustering in PCA Space



Fig:4.3.2 KMeans Clustering in PCA Space

The PCA visualization showed that **benign traffic formed a compact cluster**, while **attack traffic was more dispersed**, indicating greater behavioral variability. Despite some overlap, the clusters exhibited clear separation, demonstrating that K-Means captured meaningful patterns in the network traffic data.

4.3.3 Evaluation and Role of K-Means Clustering

This provides as a baseline to compare with Kernel PCA + K-means

4.3.3.1 Objective

The purpose of this evaluation is to assess the performance of the K-Means clustering model and understand its contribution within the overall hybrid machine learning pipeline. Although K-Means is an unsupervised learning algorithm, its performance can be evaluated by comparing cluster assignments with true labels.

4.3.3.2 Performance Summary

i. Classification Metrics

- Accuracy: 0.83 Macro F1-score: 0.68 Weighted F1-score: 0.82

Class	Precision	Recall	F1-Score
Benign (0)	0.86	0.95	0.90
Attack (1)	0.64	0.36	0.46

Table i : Classification Metrics (Unsupervised Model-K Means)

ii. Confusion Matrix Analysis

	Predicted Benign	Predicted Attack
Actual Benign	431,342	22,376
Actual Attack	70,934	40,339

Table ii : Confusion Matrix (Unsupervised Model-K Means)

The model performs very well in identifying benign traffic (high true negatives). A significant number of attack samples are misclassified as benign (high false negatives). This indicates that attack patterns are not well captured by clustering alone.

iii. ROC Curve Analysis

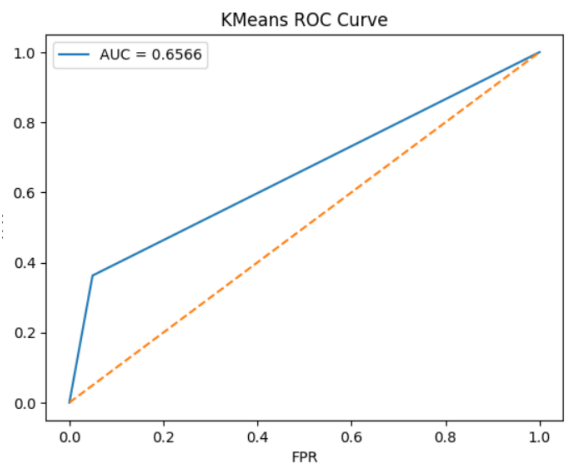


Fig iii : ROC- Curve (Unsupervised Model-K Means)

AUC Score: 0.6566.The model performs better than random guessing (AUC = 0.5). However, the discrimination capability is limited. This confirms that K-Means is not suitable as a standalone classifier.

iv. Classification report

KMeans Classification Report:				
	precision	recall	f1-score	support
0	0.86	0.95	0.90	453718
1	0.64	0.36	0.46	111273
accuracy			0.83	564991
macro avg	0.75	0.66	0.68	564991
weighted avg	0.82	0.83	0.82	564991

Fig iv : Classification Report (Unsupervised Model-K Means)

K-Means achieved an overall accuracy of 83%, performing well on benign traffic detection but showing lower recall (36%) for attack traffic, indicating difficulty in identifying malicious network activity.

v. Confusion Matrix

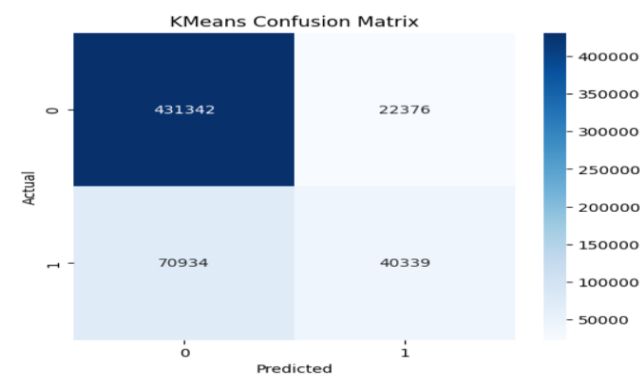


Fig v : Confusion Matrix (Unsupervised Model-K Means)

The K-Means model achieved a Silhouette Score of 0.4893, indicating moderate cluster separation, with strong benign traffic detection but a high number of attack instances misclassified as benign.

4.4 Model 2 : Unsupervised Learning Model: Kernel PCA + K-Means Clustering

4.4.1 Objective

The objective of this unsupervised learning model was to identify latent patterns in network traffic without using class labels during the clustering process. Unlike supervised learning approaches such as Random Forest, which rely on known attack labels, this model attempts to discover natural groupings within the CICIDS2017 dataset and subsequently map those clusters to benign and malicious traffic. The output of this model is later used as a meta-feature in the final ensemble model (Model 3), where the clustering predictions and cluster distance metrics are combined with supervised model outputs.

4.4.2 Methodology

4.4.2.1 Feature Selection

The top 15 most important features identified using Random Forest feature importance were selected to focus clustering on the most informative network traffic characteristics.

4.4.2.2 Dataset Preparation

A balanced dataset was created by randomly sampling 10,000 benign and 10,000 attack records, resulting in 20,000 network flows. This prevents cluster formation from being biased toward the majority benign class.

4.4.2.3 Feature Standardization

All selected features were standardized using StandardScaler to ensure each feature has a mean of 0 and standard deviation of 1, preventing features with larger values from dominating the clustering process.

4.4.2.4 Non-Linear Feature Transformation

To capture complex non-linear attack patterns, an RBF kernel was approximated using the Nystroem method ($\gamma = 0.01$, $n_components = 500$). This approach improves cluster separability while remaining computationally efficient for large datasets.

4.4.2.5 Dimensionality Reduction

PCA was applied to reduce the transformed 500-dimensional feature space to 10 principal components, reducing noise and computational complexity while retaining the most significant information.

4.4.2.6 K-Means Clustering

K-Means clustering was performed on the reduced feature space. The algorithm iteratively assigns data points to the nearest cluster centroid and updates centroids until convergence, aiming to maximize cluster separation and minimize within-cluster variance.

4.4.3 Determination of Optimal Number of Clusters

The optimal number of clusters was selected using the Elbow Method and Silhouette Analysis. Since no clear elbow point was observed, the silhouette score was used, with **k = 2** achieving the highest score (**0.5823**). Therefore, two clusters were selected.

k	Inertia
2	1891.18
3	1562.09
4	1311.55
5	1058.85
6	846.19
7	660.07

k	Silhouette Score
2	0.5823
3	0.5130
4	0.4351
5	0.5047
6	0.5261
7	0.5473

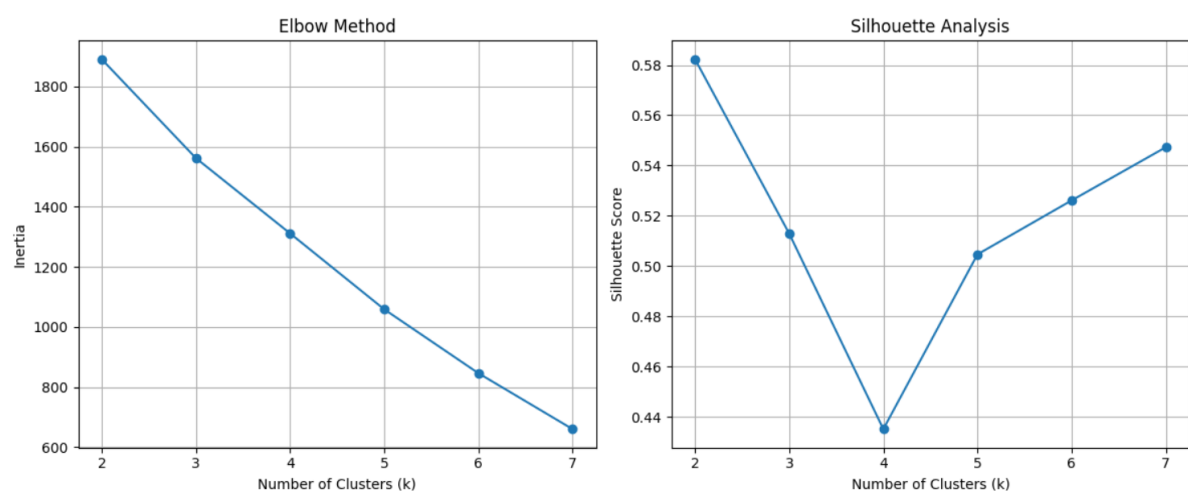


Fig: 4.4.3 Elbow Method & Silhouette Analysis (Unsupervised Model Kernel PCA+ K Means)

4.4.4 Cluster Interpretation

Clusters were labelled based on their attack rates. **Cluster 0** was assigned as **Benign**, while **Cluster 1**

Cluster	Attack Rate	Assigned Label
0	36.7%	Benign
1	93.0%	Attack

was assigned as **Attack**.

Table: 4.4.4 Cluster Interpretation (Unsupervised Model Kernel PCA+ K Means)

4.4.5 Cluster Visualization

Kernel PCA visualization showed improved separation between benign and attack traffic, with minimal overlap between clusters, indicating effective capture of non-linear patterns.

4.4.6 Classification Performance

Metric	Benign	Attack
Precision	0.87	0.77
Recall	0.97	0.44
F1 Score	0.92	0.56

Table: 4.4.6 Classification Performance (Unsupervised Model Kernel PCA+ K Means)

The Kernel PCA + K-Means model achieved 86% accuracy with strong benign traffic detection and moderate attack detection performance. The improvement in attack F1-score 0.56 indicates that the non-linear transformation enhanced cluster quality and attack identification.

4.4.7 Confusion Matrix Analysis

Actual	Predicted Benign	Predicted Attack
Benign	438,914	14,804
Attack	62,823	48,450

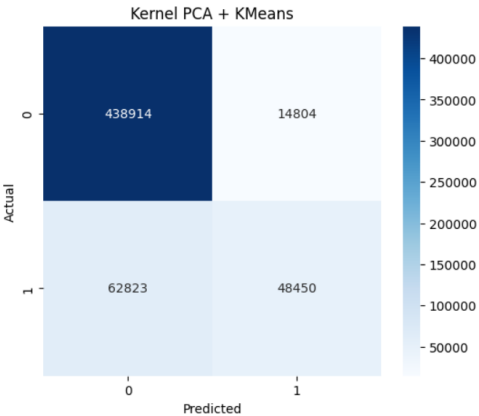


Fig: 4.4.7 Confusion Matrix Analysis (Unsupervised Model Kernel PCA+ K Means)

The confusion matrix shows that the model correctly classified most benign traffic while detecting a substantial portion of attack traffic. However, some attacks were still misclassified as benign, resulting in lower attack recall.

4.4.8 ROC-AUC Analysis

The model achieved an AUC of 0.6967, indicating moderate discrimination between benign and malicious traffic and performance significantly better than random guessing.

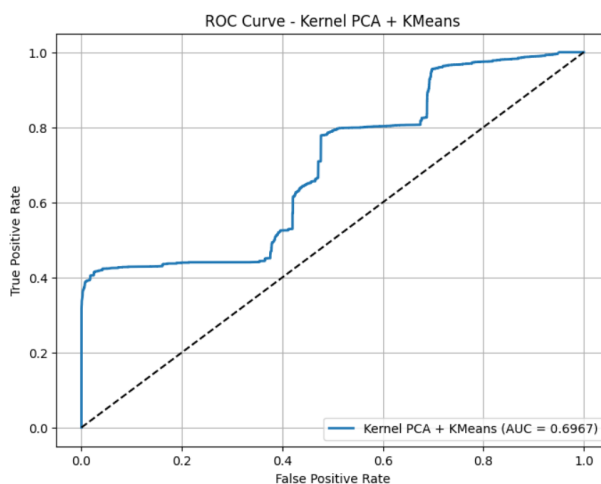


Fig: 4.4.8 ROC-AUC Analysis (Unsupervised Model Kernel PCA+ K Means)

4.4.9 Key Findings

The model achieved **86% overall accuracy** and a **Silhouette Score of 0.5823**, demonstrating well-separated clusters and reduced overlap between benign and attack traffic. Additionally, the clustering process generated meaningful meta-features that enhanced the final ensemble model. However, the model achieved only **44% attack recall**, indicating that some attack samples remained embedded within benign clusters, resulting in lower performance than the supervised learning models.

4.4.10 Role in the Hybrid Framework

The clustering model contributes additional unsupervised information to the final XGBoost ensemble through cluster assignments and distance-based features, complementing the supervised Random Forest predictions.

4.4.11 Comparative Analysis: PCA + K-Means vs Kernel PCA + K-Means

The objective was to evaluate whether a non-linear Kernel PCA transformation improves K-Means clustering performance compared to traditional PCA.

4.4.12 Clustering Quality

Kernel PCA achieved a higher Silhouette Score (0.5823 vs 0.4893), indicating better cluster cohesion and separation. The Kernel PCA approach improved the silhouette score by approximately **19%**.

4.4.13 Classification Performance

Kernel PCA improved overall accuracy, attack precision, attack recall, F1-score, and ROC-AUC compared to the PCA-based model. Kernel PCA improved overall classification accuracy by approximately **3.6%**, attack precision by **20.3%**, attack recall by **22.2%** and F1 score by **21.7%**.

Model	Accuracy	Model	Recall
PCA + K-Means	83%	PCA + K-Means	0.36
Kernel PCA + K-Means	86%	Kernel PCA + K-Means	0.44

Model	F1 Score	Model	AUC
PCA + K-Means	0.46	PCA + K-Means	0.6566
Kernel PCA + K-Means	0.56	Kernel PCA + K-Means	0.6967

Fig: 4.4.13 Classification Performance (Unsupervised Model Kernel PCA+ K Means)

4.4.14 Confusion Matrix Comparison

The Kernel PCA model detected more attack samples while simultaneously reducing false positives, demonstrating improved intrusion detection capability.

Actual	Predicted Benign	Predicted Attack
Benign	431,342	22,376
Attack	70,934	40,339

PCA + K-Means

Attack Detection Rate:

40,339 / 111,273 = 36.3%

Actual	Predicted Benign	Predicted Attack
Benign	438,914	14,804
Attack	62,823	48,450

Kernel PCA + K-Means
 Attack Detection Rate:
 $48,450 / 111,273 = 43.5\%$

Fig: 4.4.14 Confusion Matrix Comparison (Unsupervised Model Kernel PCA+ K Means)

The Kernel PCA model successfully identified **8,111 additional attack samples** that were missed by the PCA-based clustering model.

False positives reduced from: 22,376 → 14,804

Reduction achieved: $22,376 - 14,804 = 7,572$ **fewer false alarms**. This demonstrates that Kernel PCA simultaneously improved attack detection while reducing false alerts.

4.4.15 Visual Cluster Separation

Compared to traditional PCA, Kernel PCA produced clearer cluster separation, more compact cluster structures, and reduced overlap between benign and attack traffic. These visual improvements are consistent with the higher silhouette score, improved attack recall, and increased AUC achieved by the Kernel PCA + K-Means model.

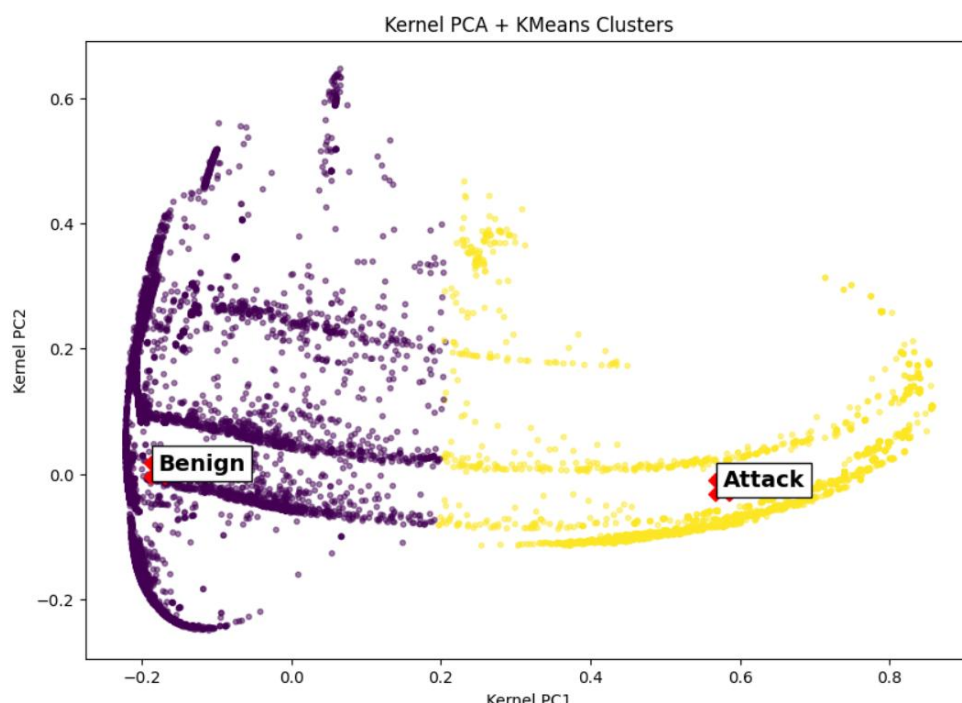


Fig: 4.4.15 Visual Cluster Separation (Unsupervised Model Kernel PCA+ K Means)

4.4.16 Overall Findings

Kernel PCA consistently outperformed traditional PCA across all evaluation metrics, producing better cluster separation and more effective attack detection. This model generates **cluster_pred**, **cluster_dist_0**, and **cluster_dist_1**, capturing latent traffic structure and cluster proximity information that enriches the final XGBoost ensemble with behavioural patterns not directly learned through supervised models.

attack_prob_Benign	attack_prob_Credential_Attack	attack_prob_DoS_DDoS	attack_prob_Malware_Compromise	attack_prob_Reconnaissance	attack_prob_Web_Attack	cluster_dist_0	cluster_dist_1	cluster_pred
1.000000	0.0	0.000000	0.0	0.000000	0.0	0.122367	0.796829	0
0.987650	0.0	0.012329	0.0	0.000021	0.0	0.142520	0.801410	0
1.000000	0.0	0.000000	0.0	0.000000	0.0	0.277138	0.743539	0
0.987650	0.0	0.012329	0.0	0.000021	0.0	0.142520	0.801410	0
0.993333	0.0	0.006667	0.0	0.000000	0.0	0.698370	0.679889	1

Fig: 4.4.16 Meta Feature Separation (Unsupervised Model Kernel PCA+ K Means)

Metric	PCA + K-Means	Kernel PCA + K-Means	Improvement
Silhouette Score	0.4893	0.5823	+19.0%
Accuracy	83%	86%	+3.6%
Attack Precision	0.64	0.77	+20.3%
Attack Recall	0.36	0.44	+22.2%
Attack F1 Score	0.46	0.56	+21.7%
AUC	0.6566	0.6967	+6.1%

Table: 4.4.16 Improvement Analysis (Unsupervised Model Kernel PCA+ K Means)

4.4.17 Conclusion

Kernel PCA significantly enhanced K-Means clustering by capturing non-linear relationships within network traffic data. The resulting model achieved better clustering quality, improved attack detection, fewer false alarms, and stronger overall performance than the PCA-based approach.

Consolidate output from Model 1 (Random Forest) , Model 2 (Random Forest Softmax) , Model 3 (Kernal PCA + K-Means) which goes to final XGBoost Model.

4.5 Model 3: Final Hybrid Ensemble Model: Integration of Supervised, Unsupervised, and Attack-Type Intelligence

4.5.1 Objective

The objective of the final hybrid model is to combine insights from multiple learning approaches into a unified intrusion detection framework. The model integrates binary intrusion predictions from Random Forest, attack-category predictions and probabilities from a multiclass Random Forest, structural information derived from Kernel PCA + K-Means clustering, and original network traffic features. These complementary features are then provided to an XGBoost classifier, which serves as the final decision-making model for detecting malicious network activity.

4.5.2 Hybrid Architecture

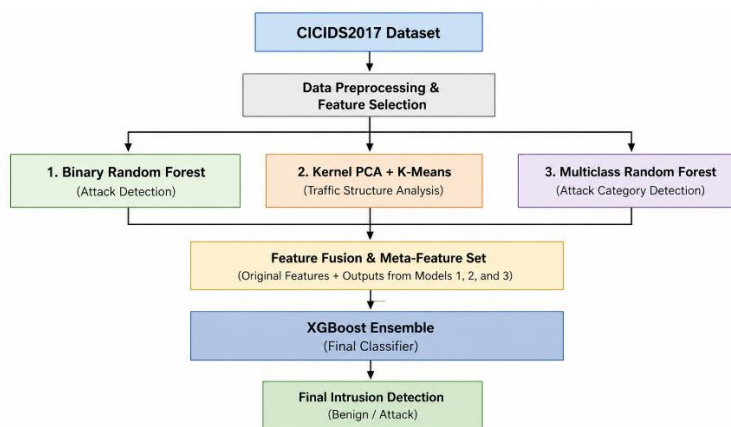


Fig: 4.5.2 Hybrid Architecture (Final Ensemble Model)

4.5.3 Feature Preparation

The final dataset was constructed using the original CICIDS2017 traffic features together with outputs generated by the preceding models. The Binary Random Forest contributed `rf_pred` and `rf_prob`, representing the predicted traffic class and attack probability. The Attack-Type Random Forest provided `attack_class_pred` along with six attack-category probability features: `attack_prob_Benign`, `attack_prob_Credential_Attack`, `attack_prob_DoS_DDoS`, `attack_prob_Malware_Compromise`, `attack_prob_Reconnaissance`, and `attack_prob_Web_Attack`. The Kernel PCA + K-Means model contributed `cluster_pred`, `cluster_dist_0`, and `cluster_dist_1`, capturing cluster membership and distance from cluster centroids. These features were combined with the original network traffic attributes to create a rich feature set for the final XGBoost ensemble model.

4.5.4 Feature Selection and Data Splitting

To preserve information from all three learning paradigms, a set of **12 mandatory features** was retained in the final dataset, including **Binary Random Forest outputs** (rf_pred, rf_prob), **Attack-Type Random Forest outputs** (attack_class_pred and six attack_prob_* features), and **Kernel PCA + K-Means outputs** (cluster_pred, cluster_dist_0, cluster_dist_1). A Random Forest model (**100 trees, random_state=42**) was then used to compute feature importance scores, and the **top 20 features** were selected. These top-ranked features were merged with the mandatory features to reduce dimensionality while preserving all ensemble signals.

The resulting feature set was split into **80% training** and **20% testing** data using a **stratified train-test split** (random_state=42), ensuring that the class distribution remained consistent across both datasets.

4.5.5 Final Classifier: XGBoost

The final ensemble model utilized XGBoost (objective="binary:logistic", eval_metric="logloss") to integrate supervised, unsupervised, and attack-category intelligence. Hyperparameters were optimized using GridSearchCV with 3-fold cross-validation and ROC-AUC scoring. A total of 16 parameter combinations (48 model fits) were evaluated across n_estimators (100, 200), max_depth (4, 6), learning_rate (0.05, 0.10), and subsample (0.8, 1.0).

Best Hyperparameters: n_estimators=200, max_depth=6, learning_rate=0.10, and subsample=0.80. This configuration provided an effective balance between predictive performance, model complexity, and generalization.

4.5.6 Hybrid Architecture Advantages

The proposed framework differs from traditional intrusion detection systems by combining **binary intrusion detection**, **attack-type classification**, and **unsupervised clustering intelligence** within a single ensemble architecture. The Binary Random Forest contributes **rf_pred** and **rf_prob**, the multiclass Random Forest provides **attack_class_pred** and **attack_prob_*** features, while Kernel PCA + K-Means contributes **cluster_pred**, **cluster_dist_0**, and **cluster_dist_1**.

This integration enables the model to simultaneously capture **network traffic characteristics**, **attack likelihood**, **attack-category confidence**, and **latent cluster structure**. By combining complementary

information from multiple learning paradigms, the ensemble improves generalization, reduces reliance on any single model, and enhances robustness against diverse attack patterns.

Final Feature Categories

Feature Group	Purpose
Original Network Features	Traffic characteristics
rf_pred, rf_prob	Binary attack intelligence
attack_class_pred, attack_prob_*	Attack-category intelligence
cluster_pred, cluster_dist_*	Clustering-based structural information

Conclusion

The final model integrates **Random Forest binary predictions, attack-type probabilities, Kernel PCA + K-Means clustering features**, and **original network traffic attributes** into a unified XGBoost ensemble. By learning from these complementary information sources, the proposed hybrid architecture provides a richer representation of network behavior and delivers improved intrusion detection performance, generalization, and robustness compared to standalone approaches.

4.5.7 Evaluation and Analysis

4.5.7.1 Classification Performance

					Metric	Score
Class	Precision	Recall	F1-Score	Support	Accuracy	99.79%
Benign (0)	1.00	1.00	1.00	90,744	Macro Precision	1.00
Attack (1)	1.00	0.99	0.99	22,255	Macro Recall	1.00
					Macro F1 Score	1.00
					Weighted F1 Score	1.00

Fig:4.5.7.1 Classification Report (Final Ensemble Model)

Overall Performance

The final hybrid model achieved near-perfect performance, accurately distinguishing between benign and malicious traffic with extremely high precision and recall. The results demonstrate that combining binary predictions, attack-type intelligence, and clustering-based features significantly improves intrusion detection effectiveness and robustness.

4.5.7.2 Confusion Matrix Analysis

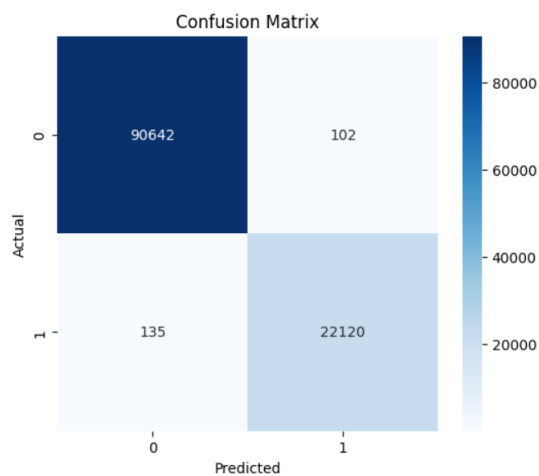


Fig:4.5.7.2 Confusion Matrix Analysis (Final Ensemble Model)

The confusion matrix demonstrates excellent class separation, with **90,642 true negatives** and **22,120 true positives** correctly classified. The model produced only **102 false positives (0.11%)** and **135 false negatives (0.61%)**, indicating a very low false alarm rate while successfully detecting the vast majority of attacks. These results highlight the model's reliability and effectiveness for intrusion detection applications.

4.5.7.3 ROC-AUC Analysis

The model achieved an AUC score of 0.9998, indicating near-perfect discriminative capability between benign and malicious traffic. The ROC curve closely follows the upper-left boundary, demonstrating extremely high sensitivity and specificity, with the model effectively distinguishing attacks from normal network activity.

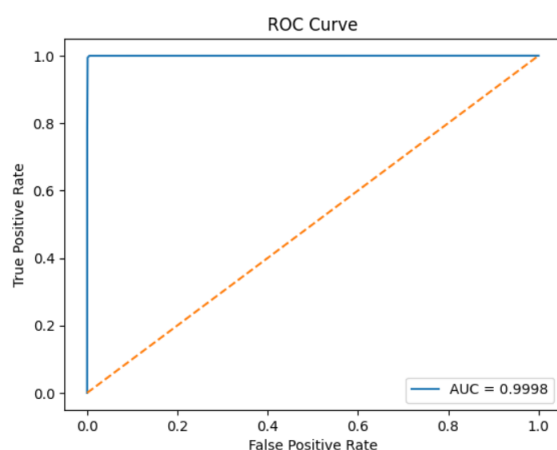


Fig:4.5.7.3 ROC- AUC Analysis (Final Ensemble Model)

4.5.7.4 Learning Curve Analysis

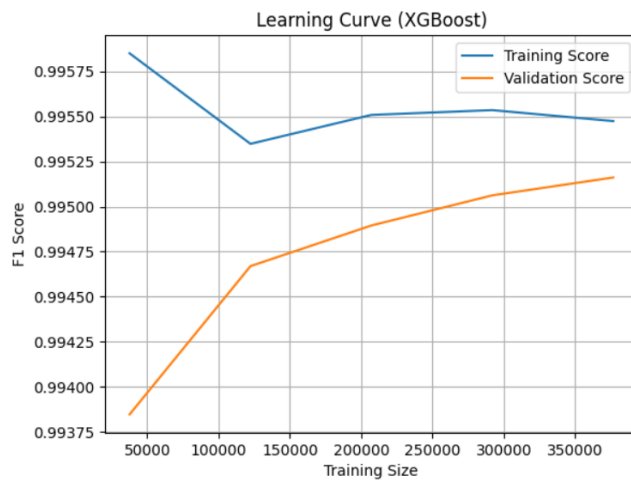


Fig:4.5.7.4 Learning Curve Analysis (Final Ensemble Model)

The learning curve demonstrates strong and stable model performance, with training F1 scores ranging from 0.9954 to 0.9958 and validation F1 scores ranging from 0.9938 to 0.9952. The small gap between the curves indicates minimal overfitting, while the steady improvement in validation performance with increasing training data suggests good generalization. Overall, the model exhibits stable learning behaviour and sufficient complexity to effectively capture network intrusion patterns.

4.5.7.5 Feature Importance Analysis

The feature importance results show that `attack_prob_Benign` and `rf_prob` were the most influential predictors, highlighting the strong contribution of both the attack-type and binary Random Forest models. Clustering features (`cluster_dist_0` and `cluster_dist_1`) also ranked among the important features, demonstrating that the Kernel PCA + K-Means model provided valuable structural information. Additionally, several original network traffic features, including Packet Length Std, Packet Length Mean, Max Packet Length, Average Packet Size, Subflow Fwd Bytes, and Bwd Packet Length Std, remained highly relevant, indicating that the ensemble effectively leverages both learned model outputs and raw traffic characteristics.

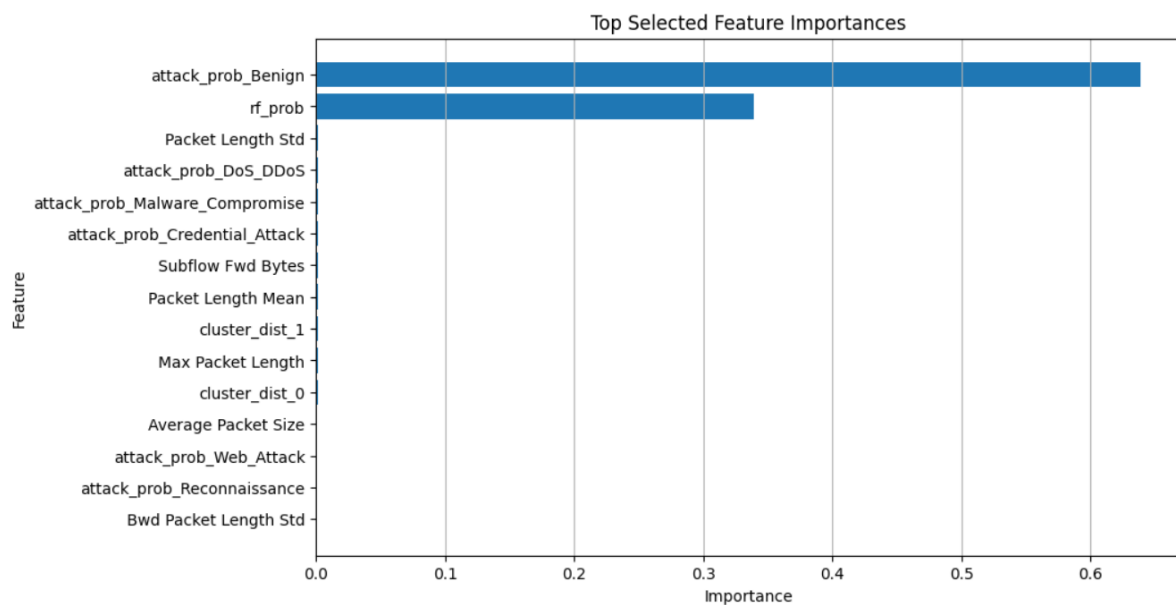


Fig:4.5.7.5 Feature Importance Analysis (Final Ensemble Model)

5. OVERALL ASSESSMENT AND CONCLUSION

The feature importance analysis confirms successful integration of all components of the hybrid framework, including **binary Random Forest outputs (rf_prob)**, **attack-type probabilities (attack_prob_*)**, **Kernel PCA + K-Means clustering features (cluster_dist_0, cluster_dist_1)**, and **original network traffic features** such as Packet Length statistics and Subflow Fwd Bytes. This demonstrates that the model effectively combines **behavioral, attack-type, structural, and raw traffic information** within a unified decision-making process.

The final XGBoost ensemble achieved **99.79% accuracy**, **0.9998 ROC-AUC**, **1.00 attack precision**, **0.99 attack recall**, and **0.99 attack F1-score**, while maintaining a **0.11% false positive rate** and **0.61% false negative rate**. Learning curve analysis further indicated strong generalization and minimal overfitting. These results demonstrate that the proposed hybrid architecture provides a highly accurate, robust, and scalable solution for network intrusion detection.

6. DEPLOYMENT STRATEGY

The proposed hybrid intrusion detection system is designed for **real-time deployment**, where network flows are captured, pre-processed, and analyzed using the **Binary Random Forest**, **Kernel PCA + K-Means**, and **Multiclass Random Forest** models. The generated meta-features are then aggregated and processed by the **XGBoost ensemble** to produce the final intrusion and attack-category prediction.

The system can be deployed as a **REST API** using **FastAPI** or **Flask** on on-premise servers, virtual machines, Docker containers, or cloud platforms such as AWS, Azure, and GCP. Due to the computational efficiency of Random Forest, XGBoost, and K-Means inference, the solution offers **low latency, moderate deployment cost, and near real-time performance** without requiring GPU resources. Periodic monitoring and retraining are recommended to address concept drift and maintain detection accuracy.

7. LIMITATIONS AND FUTURE SCOPE

Despite achieving excellent performance, the proposed hybrid framework has several limitations. The model relies heavily on meta-features generated by upstream models, introducing potential risks of data leakage and error propagation. The ensemble architecture is also computationally complex and less interpretable than standalone models. In addition, severe class imbalance affected minority attack categories such as **Web_Attack** and **Malware_Compromise**, while the model's generalization to zero-day attacks, emerging threats, and environments beyond the **CICIDS2017** dataset remains unverified. Furthermore, the near-perfect performance (**Accuracy $\approx$ 99.8%, AUC $\approx$ 0.9998**) requires validation on external datasets to confirm real-world robustness.

Future work will focus on implementing a leakage-free stacking framework, evaluating advanced ensemble methods such as **LightGBM** and **CatBoost**, and exploring deep learning approaches including **Autoencoders**, **LSTMs**, and **Transformers**. Additional research will target real-time deployment, explainable AI techniques, improved detection of rare attack categories, concept drift adaptation, graph-based network

8. REFERENCES

- [1] I. Sharafaldin, A. H. Lashkari, and A. A. Ghorbani, "Toward Generating a New Intrusion Detection Dataset and Intrusion Traffic Characterization," *Proceedings of the 4th International Conference on Information Systems Security and Privacy (ICISSP)*, pp. 108–116, 2018.
- [2] S. Chavan, P. Patil, and R. Deshmukh, "Intrusion Detection on CICIDS2017 Using Random Forest," *International Journal of Computer Applications (IJCA)*, vol. 187, no. 12, pp. 15–22, 2025.
- [3] M. Alzahrani, A. Alshamrani, and M. Aldossary, "Hybrid Clustering-Based Intrusion Detection System Using Machine Learning Techniques," *arXiv preprint arXiv:2208.09711*, 2022.

- [4] X. Zhang, Y. Wang, and H. Liu, "Network Intrusion Detection Based on XGBoost Machine Learning Algorithm," *Computers & Security*, vol. 138, 2024.
- [5] A. Verma, S. Gupta, and R. Kumar, "Stacking-Based Ensemble Learning for Network Intrusion Detection," *arXiv preprint arXiv:2003.08585*, 2020.
- [6] S. Aljawarneh, M. Aldwairi, and M. B. Yassein, "PCA-Based Feature Reduction for Efficient Intrusion Detection Systems," *Applied Sciences (MDPI)*, vol. 13, no. 4, 2023.
- [7] D. Novakovic, N. Jovanovic, and M. Radovanovic, "Hybrid Feature Selection Techniques for Intrusion Detection Systems," *Informatica*, vol. 47, no. 3, pp. 401–412, 2023.
- [8] B. Schölkopf, A. Smola, and K. R. Müller, "Kernel Principal Component Analysis," *Proceedings of the 7th International Conference on Artificial Neural Networks (ICANN)*, pp. 583–588, 1997.
- [9] M. A. Ferrag, L. Maglaras, H. Janicke, J. Jiang, and M. Shu, "A Systematic Review of Data Mining and Machine Learning for Cyber Security Intrusion Detection," *IEEE Communications Surveys & Tutorials*, vol. 24, no. 1, pp. 442–485, 2022.
- [10] T. G. Dietterich, "Ensemble Methods in Machine Learning," *Multiple Classifier Systems*, Lecture Notes in Computer Science, vol. 1857, Springer, pp. 1–15, 2000.